

Cartesian Plane, Point Drawing, and DDA Line Drawing

1. Why This Topic Matters

In computer graphics, every shape is made from pixels.

A line is not drawn as one perfect math object on screen.

It is drawn as many small points (pixels).

2. Cartesian Plane Basics

A Cartesian plane has:

- X-axis (horizontal)
- Y-axis (vertical)
- Origin at (0, 0)

The 4 regions (quadrants)

1. Quadrant I: $x > 0, y > 0$ (top-right)
2. Quadrant II: $x < 0, y > 0$ (top-left)
3. Quadrant III: $x < 0, y < 0$ (bottom-left)
4. Quadrant IV: $x > 0, y < 0$ (bottom-right)

Quick sign table:

Quadrant	x sign	y sign
I	+	+
II	-	+
III	-	-
IV	+	-

3. Cartesian vs Screen Coordinates

Math class (Cartesian):

- Origin is at center (in this tutorial)
- y goes upward

Screen (Pygame):

- Origin is top-left corner of the window
- y goes downward

So we convert coordinates:

- Screen to Cartesian:
 - $cx = sx - width/2$
 - $cy = height/2 - sy$
- Cartesian to Screen:
 - $sx = width/2 + cx$
 - $sy = height/2 - cy$

These are used in both files so students can think in normal math coordinates.

What this file

1. Draw a grid and axes for reference.
2. Convert mouse click from screen space to Cartesian space.
3. Plot points and show coordinate labels.
4. Store points in a list and redraw each frame.

Controls

- Mouse click: plot point
 - C: clear all points
 - ESC: quit
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5. Part B: DDA Line Drawing

What is DDA?

DDA means Digital Differential Analyzer.

It draws a line by moving in very small equal steps from start point to end point.

Core formulas

Given two points (x_0, y_0) and (x_1, y_1) :

- $dx = x_1 - x_0$
- $dy = y_1 - y_0$
- $steps = \max(abs(dx), abs(dy))$
- $x_inc = dx / steps$
- $y_inc = dy / steps$

Then:

1. Start from $(x, y) = (x_0, y_0)$
 2. Plot $round(x), round(y)$
 3. Add increments each step
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Why $\max(\text{abs}(\text{dx}), \text{abs}(\text{dy}))$?

Because we need enough steps in the dominant direction.
This avoids gaps in steep lines.

Special case

If start point = end point, plot one pixel only.

6. Simple Example

Line from (2, 1) to (8, 4):

- $\text{dx} = 6$
- $\text{dy} = 3$
- $\text{steps} = 6$
- $\text{x_inc} = 1$
- $\text{y_inc} = 0.5$

Generated float points:

- (2.0, 1.0)
- (3.0, 1.5)
- (4.0, 2.0)
- (5.0, 2.5)
- (6.0, 3.0)
- (7.0, 3.5)
- (8.0, 4.0)

After rounding:

- (2,1), (3,2), (4,2), (5,2), (6,3), (7,4), (8,4)

This is the pixel line shown on screen.

7. Controls in DDA

- Click 2 points: draw line
 - I: input mode (type points)
 - Enter: draw typed line
 - D: debug mode on/off
 - SPACE: next debug step
 - C: clear all
 - ESC: quit
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8. Why Debug Mode Is Great

The debug mode in `DDA_line_draw.py` shows:

- `dx, dy, steps`
 - `x_inc, y_inc`
 - current float `(x, y)`
 - rounded pixel at each step
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Quick Practice Questions

1. In which quadrant is point `(-5, 7)`?
2. In which quadrant is point `(4, -3)`?
3. For line `(1,1)` to `(5,3)`, find `dx, dy, steps`.
4. Why do we round `(x, y)` in DDA?

Answers:

1. Quadrant II
 2. Quadrant IV
 3. `dx=4, dy=2, steps=4`
 4. Because screen pixels are integer locations.
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Key Takeaway

- Cartesian plane helps students reason with math.
- Point plotting is the base of all raster graphics.
- DDA is an easy first line algorithm for learning pixel-based drawing.